

1) A final-year engineering student uses Nmap to scan the network of a local software company to identify open ports and running services. The student does not obtain Permission from the company. Before performing the scan. Although no systems are compromised and no data is stolen, the company's security team detects the activity and reports it to the cybercrime cell. As a cybersecurity professional, analyze whether the student's actions are ethical and legal. Discuss the relevant Provisions of the Information Technology (IT) Act, 2000, identify the Possible legal consequences, and explain the correct Procedure the Student should have followed Before conducting the Security Assessment.

Ethical Analysis:

- i) The student's action is unethical Because Permission was not obtained.
- ii) Ethical hacking requires written authorization Before any security testing.
- iii) Even though a damage occurred, unauthorized scanning violates Professional ethics.

Legal Analysis (IT Act, 2000):

- i) Section 43: unauthorized access or attempting to access a Computer / network may result in liability for compensation.
- ii) Section 66: If the act is done dishonestly or fraudulently, it becomes a criminal offense punishable.

with imprisonment and/or fine.

iii) The security team reporting the incident to the cybercrime cell is legally justified.

Possible legal consequences:

- i) cybercrime investigation.
- ii) Civil liability and compensation.
- iii) Criminal Prosecution if malicious intent is established.
- iv) Academic or professional disciplinary action.

Correct Procedure:

- i) obtain written Permission from the organization.
- ii) Define the scope of testing.
- iii) Sign legal Agreements (Rules of Engagement / NDA).
- iv) Conduct only approved tests.
- v) Report findings confidentially with remediation suggestions.

1) A cybersecurity researcher discovers a SQL injection vulnerability in a popular e-commerce website that could allow attackers to access customer information. Instead of privately informing the organization, the researcher immediately publishes the vulnerability details and Proof-of-concept code on social media before the company can fix the issue, cybercriminals, Analyze the incident from both ethical and legal perspectives. Explain the concept responsible of vulnerability disclosure, discuss the risks of premature public disclosure, and recommend the appropriate steps the researcher should have taken from ethical perspective.

i) The researcher acted unethically By publicly

- 3) An employee of a financial organization receives an email that appears to be from the company's IT support team requesting immediate password verification. Believing the email to be genuine, the employee submits login credentials through the provided link. The attacker uses these credentials to gain unauthorized access to confidential company data. Examine this case and identify the type of cyberattack involved. Discuss the legal implications for the attacker under the Information Technology (IT) Act, explain the responsibilities of both the organization and the employee, and recommend effective technical and organizational measures to prevent similar attacks in the future.

Type of cyberattack:

Phishing attack (credential harvesting / social engineering)

Legal Implications (IT Act, 2008);

- i) Section 43: unauthorized access and data theft
- ii) Section 66: computer-related offenses
- iii) Section 66C: Identity theft By using stolen credentials
- iv) Section 66D: cheating By Personation using computer resources.

Responsibilities of the organization:

- i) Conduct cybersecurity awareness Training.
- ii) Deploy email filtering and anti-phishing solutions.
- iii) Implement multi-factor Authentication (MFA)
- iv) Monitor suspicious login attempts.
- v) Establish an incident response plan.

Responsibilities of the Employee:

- i) verify suspicious emails.

revealing the vulnerability Before informing the company.

ii) This exposed users to unnecessary risk.

Legal Perspective:

If the disclosure leads to unauthorized exploitation or data compromise, legal action may arise under:

- i) Section 43: Damage caused to computer resources
- ii) Section 66: Computer-related offenses if dishonest or malicious intent exists.
- iii) Attackers exploiting the vulnerability are clearly liable under the IT Act.

Responsible vulnerability Disclosure:

- i) Privately notify the affected organization.
- ii) Provide Technical details confidentially.
- iii) Allow sufficient time (Typically 60-90 days) for fixing the issue.

Risks of Premature Disclosure:

- i) Increased cyberattacks
- ii) customer data theft
- iii) Loss of Public trust
- iv) Legal complications

Recommended Steps:

- i) verify the vulnerability
- ii) Avoid releasing exploit code
- iii) cooperate during remediation
- iv) Report it Privately to the company
- v) Publicly disclose only after the issue is fixed.

- i) Avoid clicking unknown links.
- ii) Report phishing attempts immediately.
- iii) Follow company and security policies.

Preventive Measures:

Technical Measures:

- i) Multi-Factor Authentication (MFA)
- ii) Spam and phishing filters
- iii) Email authentication (SPF, DKIM, DMARC)
- iv) Endpoint Security
- v) Password managers.

Organizational Measures:

- i) Regular cybersecurity awareness training.
- ii) Simulated phishing exercises.
- iii) Strong password policy
- iv) Periodic security audits

A cybersecurity student discovers a serious security vulnerability in the online banking portal of a private bank. The bank does not have a Public Bug Bounty or vulnerability disclosure program. To demonstrate the severity of the issue, the student exploits the vulnerability and retrieves a small amount of customer information. Before report the findings to the bank, critically evaluate whether the student's actions are ethical and lawful. Differentiate between ethical hacking and authorized hacking, discuss the applicable provisions of the Information Technology (IT) act, and explain the proper process for reporting vulnerabilities when an organization has no official disclosure policy.

Ethical Evaluation:

- i) Although the student's intention was good, retrieving customer information without authorization is unethical.
- ii) Ethical hacking always requires explicit permission.

Lawful Evaluation:

- i) The action is not lawful because the student exceeded authorized access.
- ii) Accessing customer information without permission violates the IT Act.

Ethical Hacking vs Unauthorized Hacking.

Ethical Hacking	Unauthorized Hacking
Written Permission obtained	No Permission
Legal	Illegal
Defined Scope	No authorized scope
Reports findings responsibly	May access confidential data
Improves security	violates privacy and law.

Applicable IT Act Provisions:

- i) Section 43: unauthorized access and data extraction
- ii) Section 66: Computer-related offenses committed dishonestly or fraudulently.

Proper vulnerability Reporting Process (No Bug Bounty Available):

- i) Do not exploit beyond confirming the vulnerability
- ii) Avoid accessing customer data
- iii) Provide Technical detail and remediation suggestions
- iv) Maintain confidentiality until the issue is fixed

Conclusion: The student had good intentions, But good intentions do not make unauthorized access legal. Security Testing must always be conducted with Prior authorization and responsible disclosure Practices.